



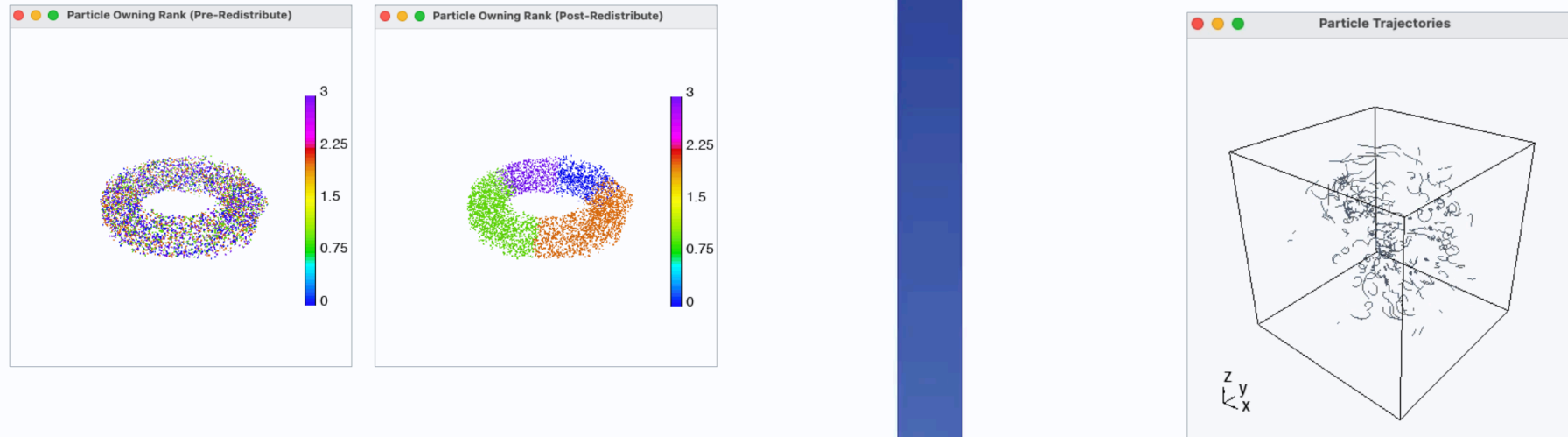
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Particle-in-cell in MFEM

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MFEM Workshop 2026

Particle tracing in MFEM



Particle Tracking in MFEM

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MFEM Workshop 2025

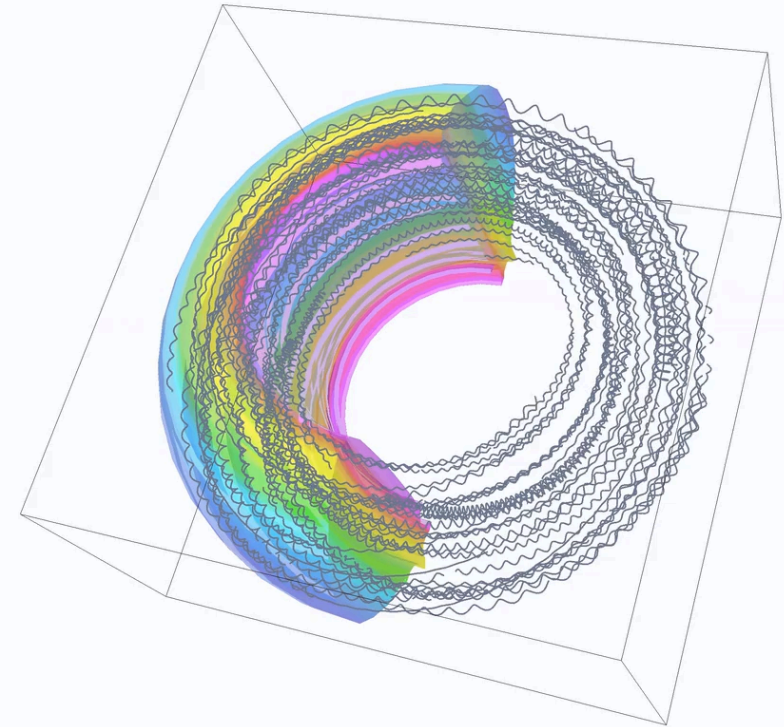
Particle tracing in MFEM

Known field $\rightarrow$ particle trajectories

Particles move through a field represented as an MFEM grid function.

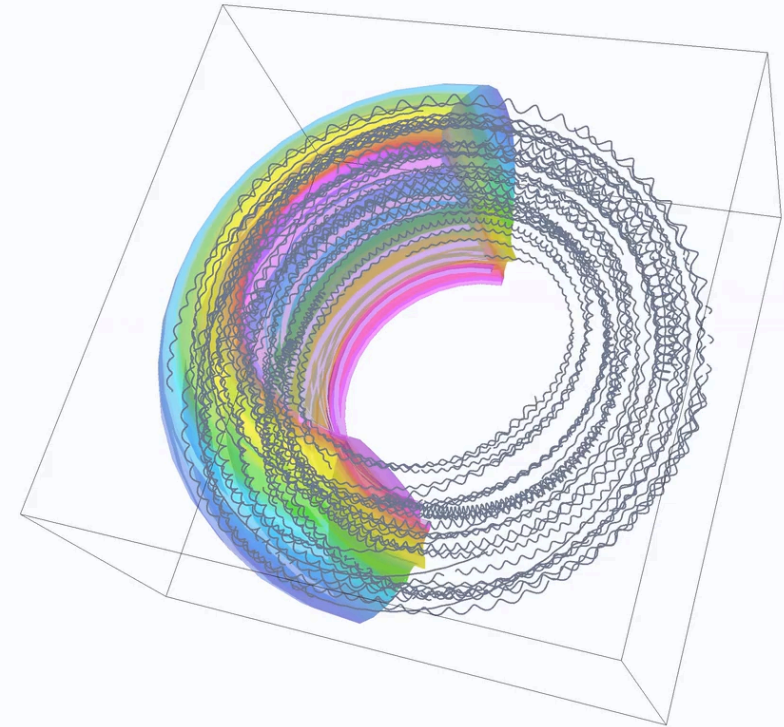
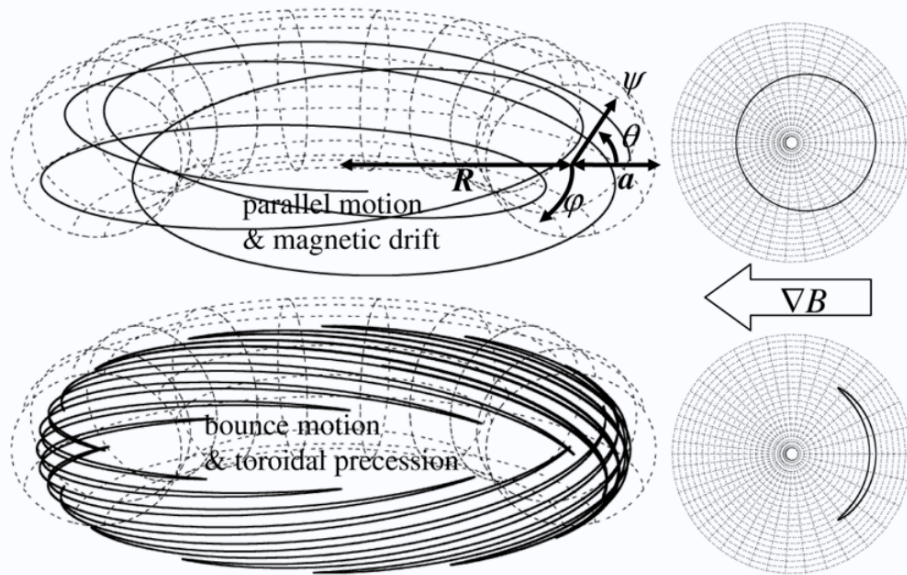
Particle tracing provides

- **Particle class** with $\mathbf{x}$, $\mathbf{p}$, m and q etc.
- **Localization** of particles in the mesh
- **Interpolation** of fields at particle positions
- **Boris** push for particle $\mathbf{x}$ and $\mathbf{p}$ updates
- **Redistribution** of particles that cross subdomain boundaries



Tokamak magnetic field generated from: R. Zhang, G. Wimmer, Q. Tang. Structure-preserving transfer of Grad-Shafranov equilibria to magnetohydrodynamic solvers, Journal of Computational Physics, 568:115369, 2027.

Particle tracing in MFEM



$\begin{matrix} y \\ z \\ x \end{matrix}$

X. Garbet, Y. Idomura, L. Villard, T. H. Watanabe. Gyrokinetic simulations of turbulent transport, Nuclear Fusion, 50:043002, 2010.

Tokamak magnetic field generated from: R. Zhang, G. Wimmer, Q. Tang. Structure-preserving transfer of Grad-Shafranov equilibria to magnetohydrodynamic solvers, Journal of Computational Physics, 568:115369, 2027.

The natural next step: particle-in-cell

particles $\xrightarrow{\text{deposit charge}}$ field $\xrightarrow{\text{push}}$ particles

Repeat this feedback loop at every time step.

Governing Equations

Consider the Vlasov-Poisson equation:

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{x}} f + \frac{e}{m} \mathbf{E} \cdot \nabla_{\mathbf{v}} f = 0.$$

Distribution function approximated by macro-particles:

$$f \approx \sum_{p=1}^{N_p} \delta(\mathbf{x} - \mathbf{x}_p) \delta(\mathbf{v} - \mathbf{v}_p).$$

Integrate over velocity to get the charge density:

$$\rho(\mathbf{x}) = e \int f(\mathbf{x}, \mathbf{v}) d\mathbf{v} = e \sum_{p=1}^{N_p} \delta(\mathbf{x} - \mathbf{x}_p).$$

Governing Equations

Poisson's equation $-\epsilon_0 \Delta \phi(\mathbf{x}) = \rho(\mathbf{x}) - \rho_0$ yields

$$-\epsilon_0 \Delta \phi(\mathbf{x}) = e \sum_{p=1}^{N_p} \delta(\mathbf{x} - \mathbf{x}_p) - en_0.$$

For a test function $\varphi \in H^1(\Omega)$, the finite element solve is

$$\epsilon_0 \langle \nabla \varphi, \nabla \phi \rangle = e \sum_{p=1}^{N_p} \varphi(\mathbf{x}_p) - en_0 \int_{\Omega} \varphi.$$

OrthoSolver is needed to remove the constant nullspace of the periodic Poisson operator and ensure a unique solution for the potential ϕ .

Governing Equations

The electric field satisfies $\mathbf{E}(\mathbf{x}) = -\nabla\phi(\mathbf{x})$, so for a test function $\mathbf{v} \in H(\text{curl}; \Omega)$, we have

$$\langle \mathbf{v}, \mathbf{E} + \nabla\phi \rangle = 0.$$

The discrete de Rham complex gives a compatible gradient,

$$\begin{array}{ccc} H^1(\Omega) & \xrightarrow{\nabla} & H(\text{curl}; \Omega) \\ \downarrow & & \downarrow \\ CG & \xrightarrow{\nabla_h} & ND \end{array}$$

following which total energy can be shown to be conserved in the discrete PIC system.

One PIC time step in MFEM

- 1. Deposit charge:** For $\varphi \in H^1(\Omega)$, assemble $e \sum_p \varphi(\mathbf{x}_p)$.
- 2. Solve Poisson:** Solve $\epsilon_0 \langle \nabla \varphi, \nabla \phi \rangle = e \sum_p \varphi(\mathbf{x}_p) - en_0 \int_{\Omega} \varphi$.
- 3. Compute field:** $\langle \mathbf{v}, \mathbf{E} + \nabla \phi \rangle = 0$.
- 4. Gather field:** interpolate $\mathbf{E}$ at each $\mathbf{x}_p$.
- 5. Leapfrog push:**

$$\mathbf{p}_p^{t+\frac{1}{2}\Delta t} = \mathbf{p}_p^{t-\frac{1}{2}\Delta t} + e\mathbf{E}^t(\mathbf{x}_p^t)\Delta t,$$

$$\mathbf{x}_p^{t+1} = \mathbf{x}_p^t + \frac{\mathbf{p}_p^{t+\frac{1}{2}\Delta t}}{m_p}\Delta t.$$

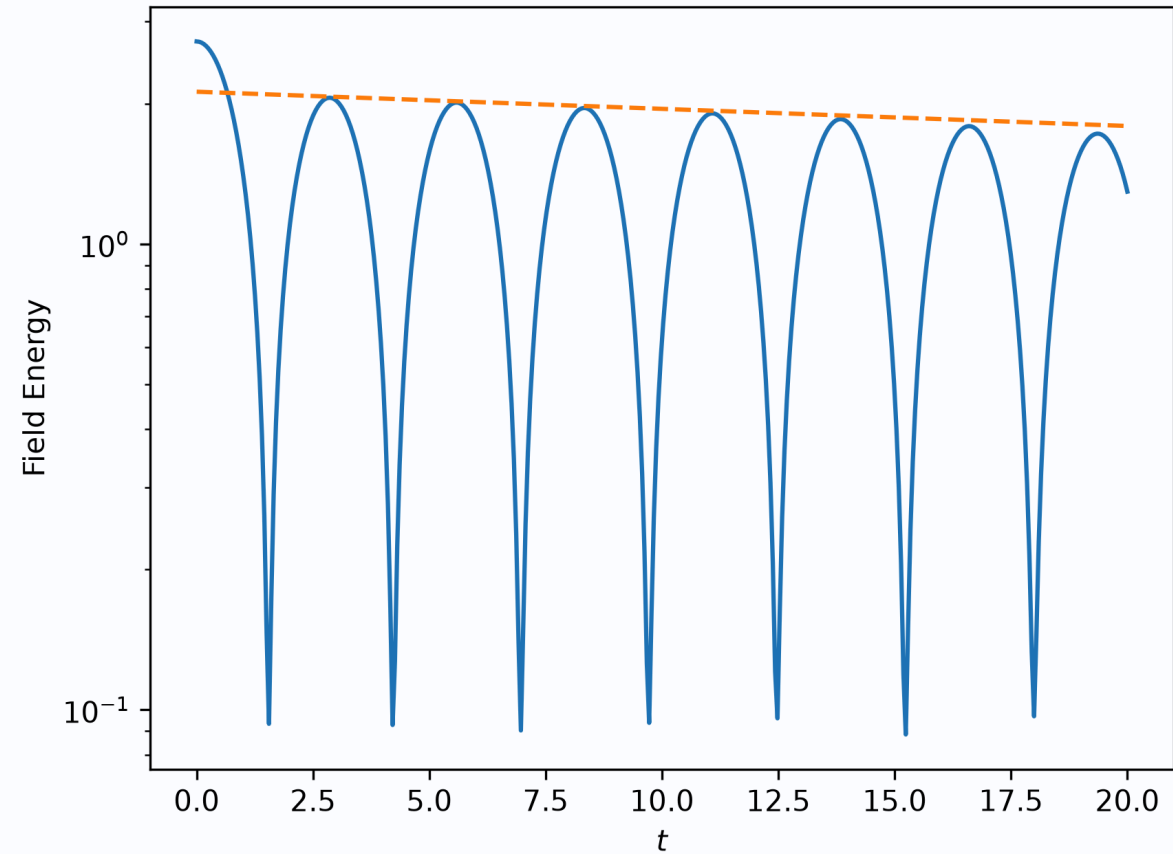
- 6. Redistribute:** transfer particles that cross MPI subdomain boundaries.

Linear Landau damping

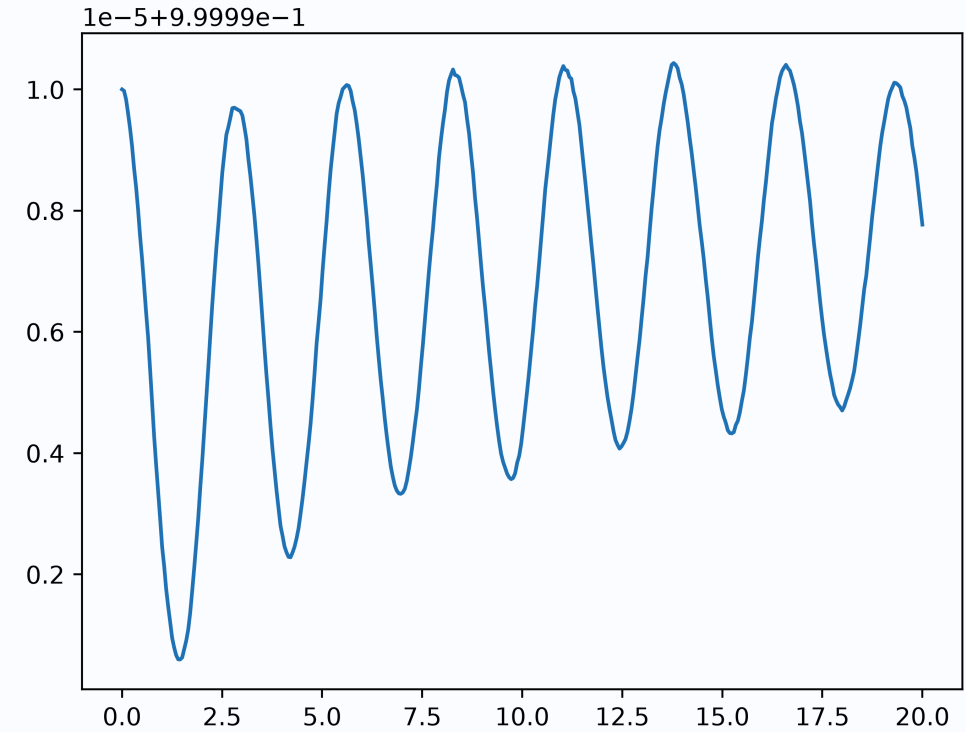
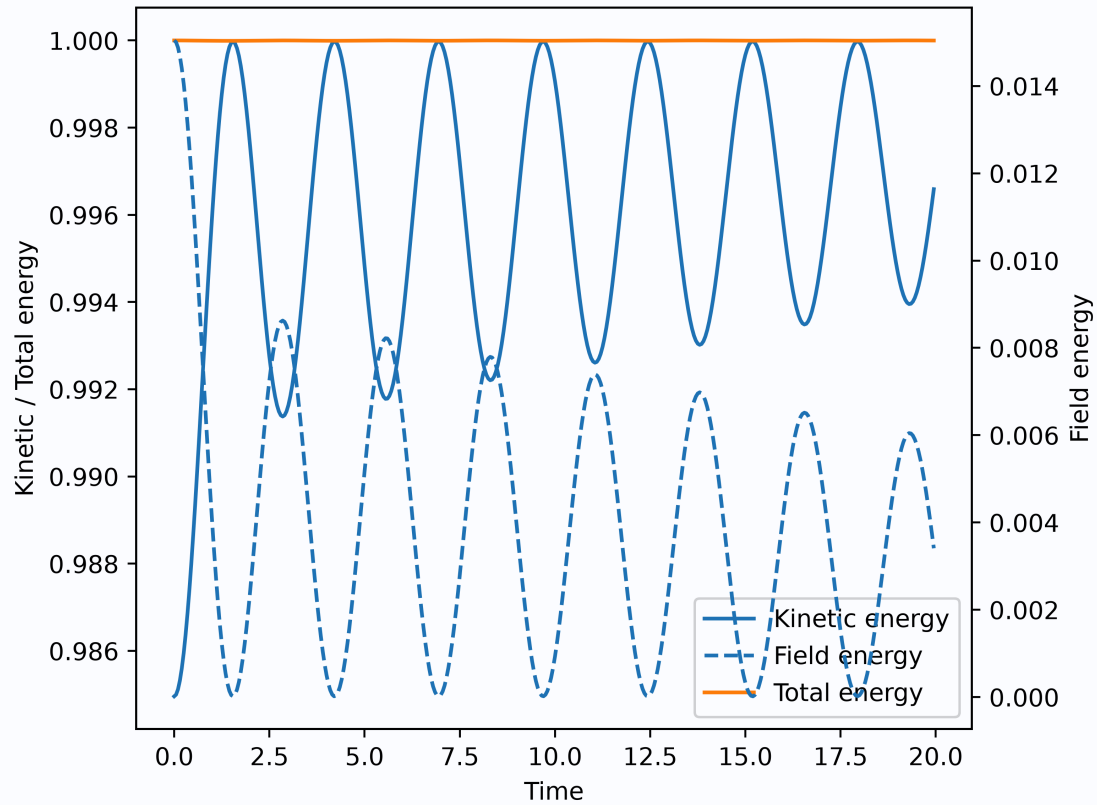
Field energy decays at a rate close to the reference rate.

256 MPI ranks · 20,000 particles per rank

64 × 64 grid · 400 steps · $\Delta t = 0.05$

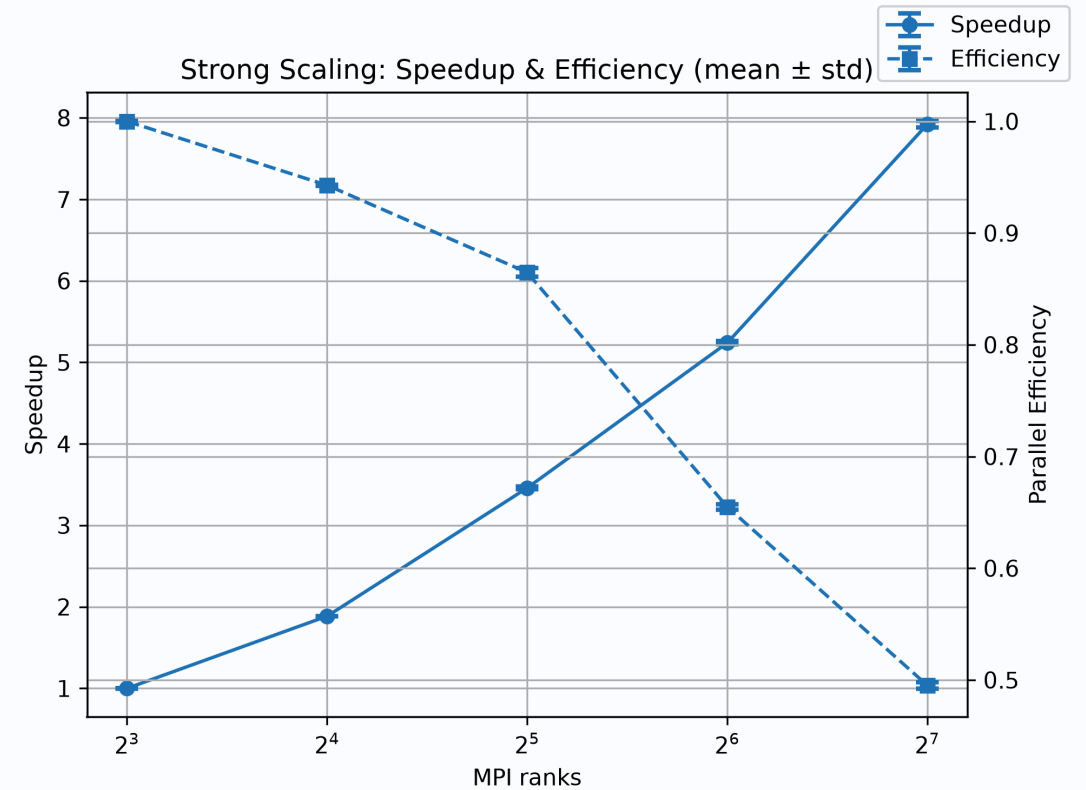
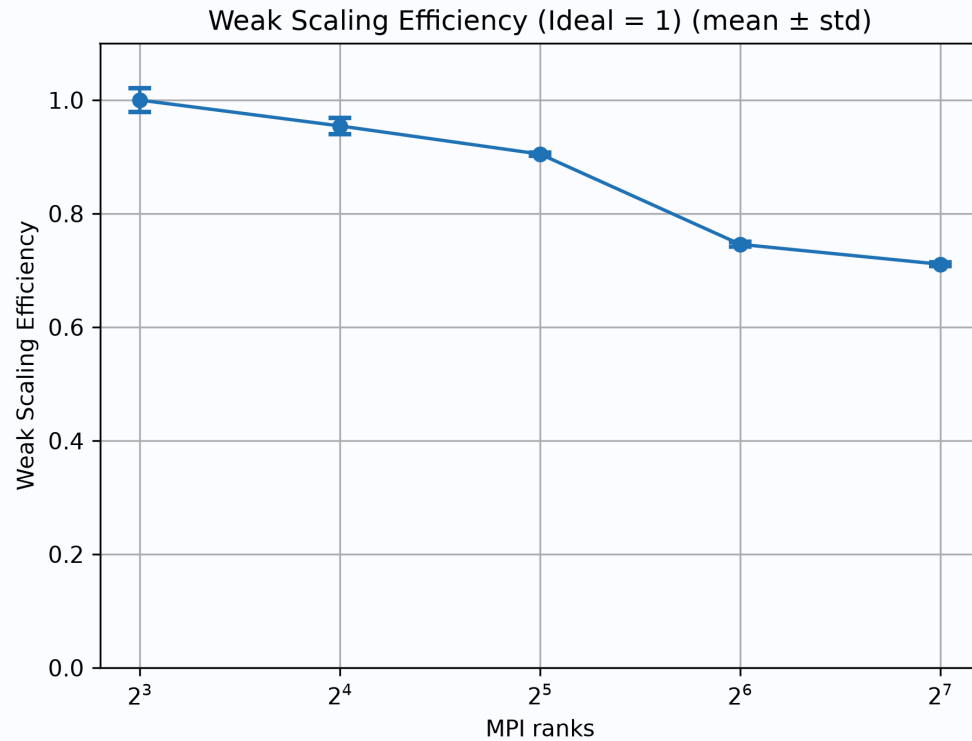


Linear Landau damping



Left: kinetic and field energy exchange. Right: total energy. (Both normalized by the initial total energy.)

Parallel scaling



Left: weak scaling, 40,960 particles per rank. Right: strong scaling, 2,621,440 particles total.

Both use 8-128 MPI ranks and a 64×64 grid. Tested on NERSC.

Summary

- Scalable electrostatic PIC implementation in MFEM
- Validated with linear Landau damping

On-going work

- Improving **scalability** on on CPU and **GPU** (See Eddy Luo and Elliot Day's poster presentation)
- Implementing a **shape function** beyond Dirac delta (See Rushan Zhang's poster presentation)

Questions?

